Land Survey

Engineering Mathematics I

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$d = \sqrt{(\Delta E)^2 + (\Delta N)^2}$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\theta = \tan^{-1}\left(\frac{\Delta E}{\Delta N}\right)$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\sin\theta = \frac{O}{H}$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\cos\theta = \frac{A}{H}$$

Section B

5. Find distance between two survey points. **(10 marks)**
6. Calculate bearing from coordinate differences. **(10 marks)**
7. Use trigonometry to calculate height. **(10 marks)**
8. Create one realistic land survey problem based on this topic and solve it fully. **(20 marks)**